Gagan Thakur

Mandi, Himachal Pradesh

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Education

National Institute of Technology, Hamirpur **Aug. 2024 – Present**
B.Tech in Electronics and Communication Engineering

Swami Vivekanand Sr. Sec. School **Apr. 2010 – Apr. 2023**
High School *Mandi, India*

Technical Skills

Languages:	Python, Rust, JavaScript, C++, C
GenAI & LLMs:	LangChain, LangGraph, LangSmith, OpenAI API, Gemini API, RAG, Vector Search
ML / DL Frameworks:	TensorFlow, PyTorch, Keras, scikit-learn, XGBoost
Data & Retrieval:	NumPy, pandas, OpenCV, FAISS, ChromaDB
MLOps & Deployment:	Git, DVC, MLflow, FastAPI, Docker, Streamlit, CI/CD, AWS

Projects

Aizen – AI-Powered Web Image Editor [GitHub] *Next.js, FFmpeg, OpenAI API, JavaScript*

- Building a web-based AI image editor where edit commands are generated by an LLM API and executed via FFmpeg, enabling natural language driven image transformations in the browser. *(In development)*

CurrCall – RAG Curriculum Chatbot [GitHub] [Live Demo] *Python, LangChain, Gemini API, RAG, FastAPI, Next.js*

- Built and deployed a full-stack RAG chatbot for the ECE department at NIT Hamirpur, enabling students to query curriculum details, course prerequisites, and academic resources via natural language.
- Architected a Python-based RAG backend using LangChain and the Gemini API, served via FastAPI, with a Next.js frontend deployed on Vercel.

Float-TCB – Ocean Data Query Chatbot [GitHub] *Python, LangChain, Vector Store, RAG*

- Built a terminal-based RAG chatbot for a hackathon to query continuously updated ocean data, featuring a real-time data ingestion pipeline, vector store retrieval, and conversational Q&A over live marine datasets.

YouTube Chat CLI [GitHub] *Python, LangChain, RAG, Vector Store, YouTube Transcript API*

- Built a Linux CLI tool that ingests YouTube video transcripts, chunks and embeds them into a vector store, and enables conversational Q&A over video content using a RAG pipeline.

Achievements

- Designed and published **26 end-to-end ML projects** (*Alphabets of Machine Learning*) spanning supervised, unsupervised, and reinforcement learning — used as a learning resource by peers.
- Solved **LeetCode Top 150** series twice, reinforcing strong foundations in data structures and algorithms.

Interests

- Digital art and creative visual design
- Literature and story writing
- Building personal projects at the intersection of AI and systems programming